

■ Augmented Security Knowledge Base ■

Team: **OS1B**

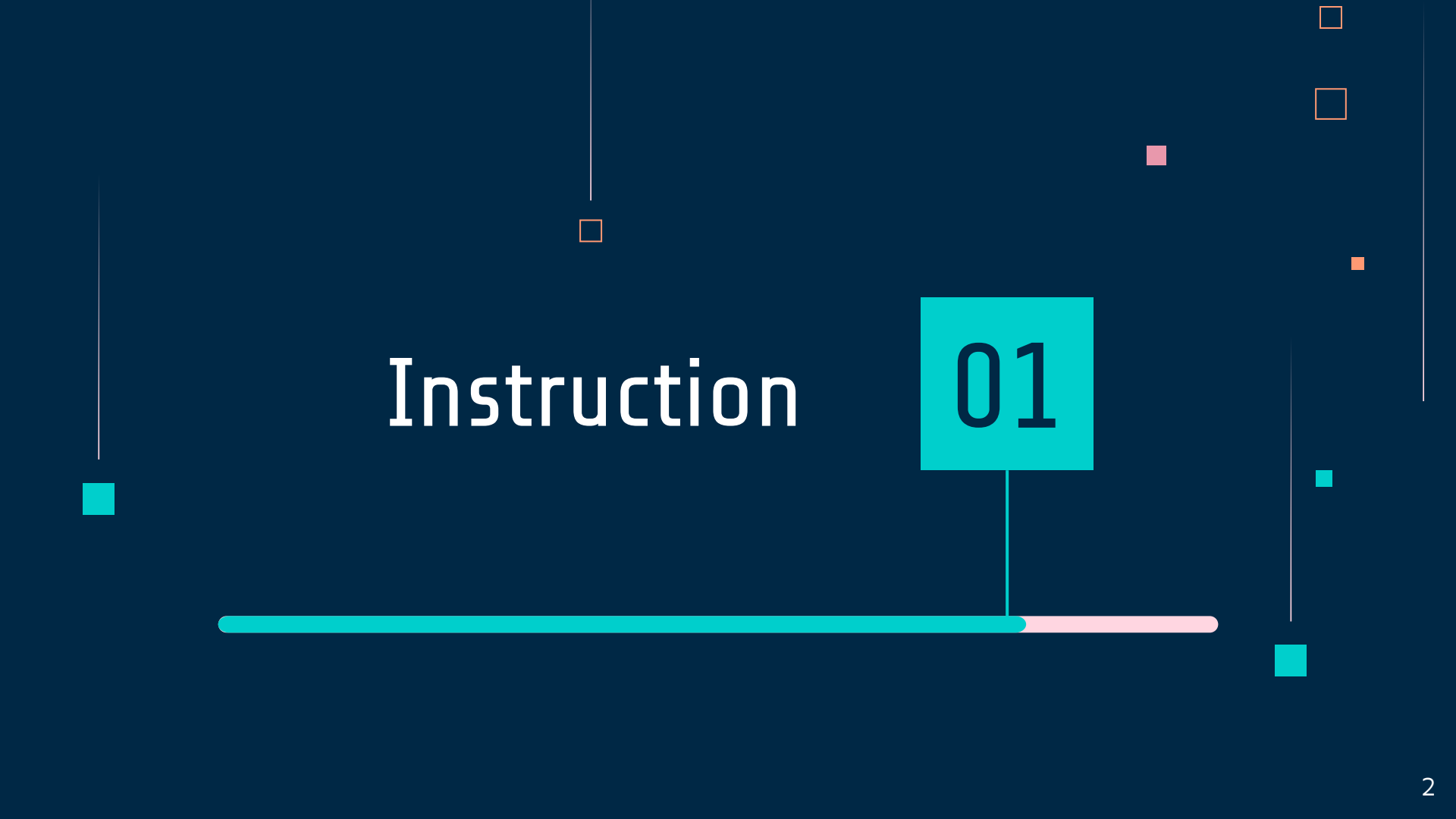
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Instruction

01

1 Introduction



COMPANIES

Many companies
suffer attacks



DEVELOPERS

Developers still lack
expertise



DEVELOPMENT

Faced real-world
challenges during the
development process.



SEARCH

Searching technology
websites takes a lot
of time

1 Introduction

■ GOAL

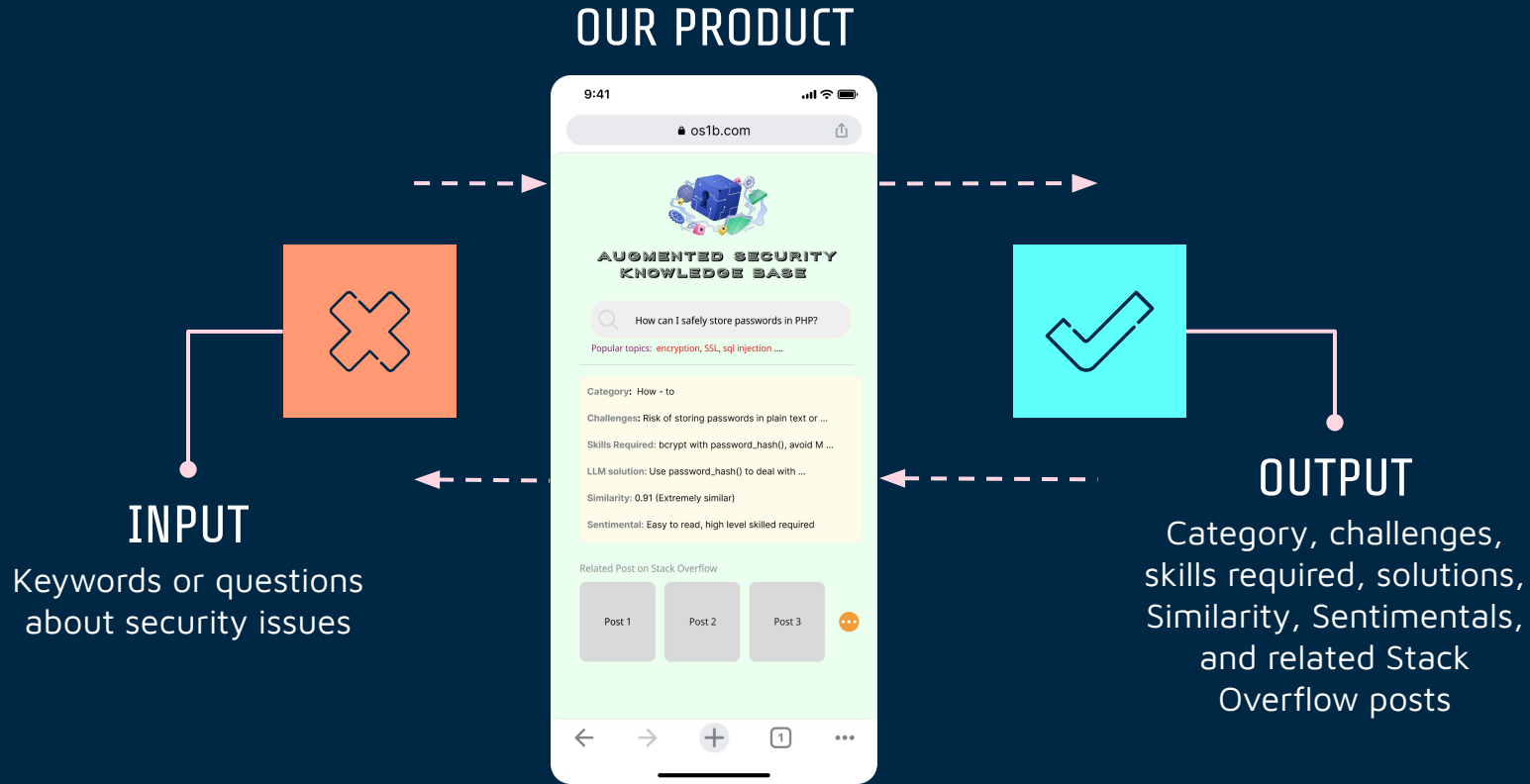
An AI-driven Augmented Knowledge Base to support developers learning security questions

■ DOMAIN SPECIFIED DATASET

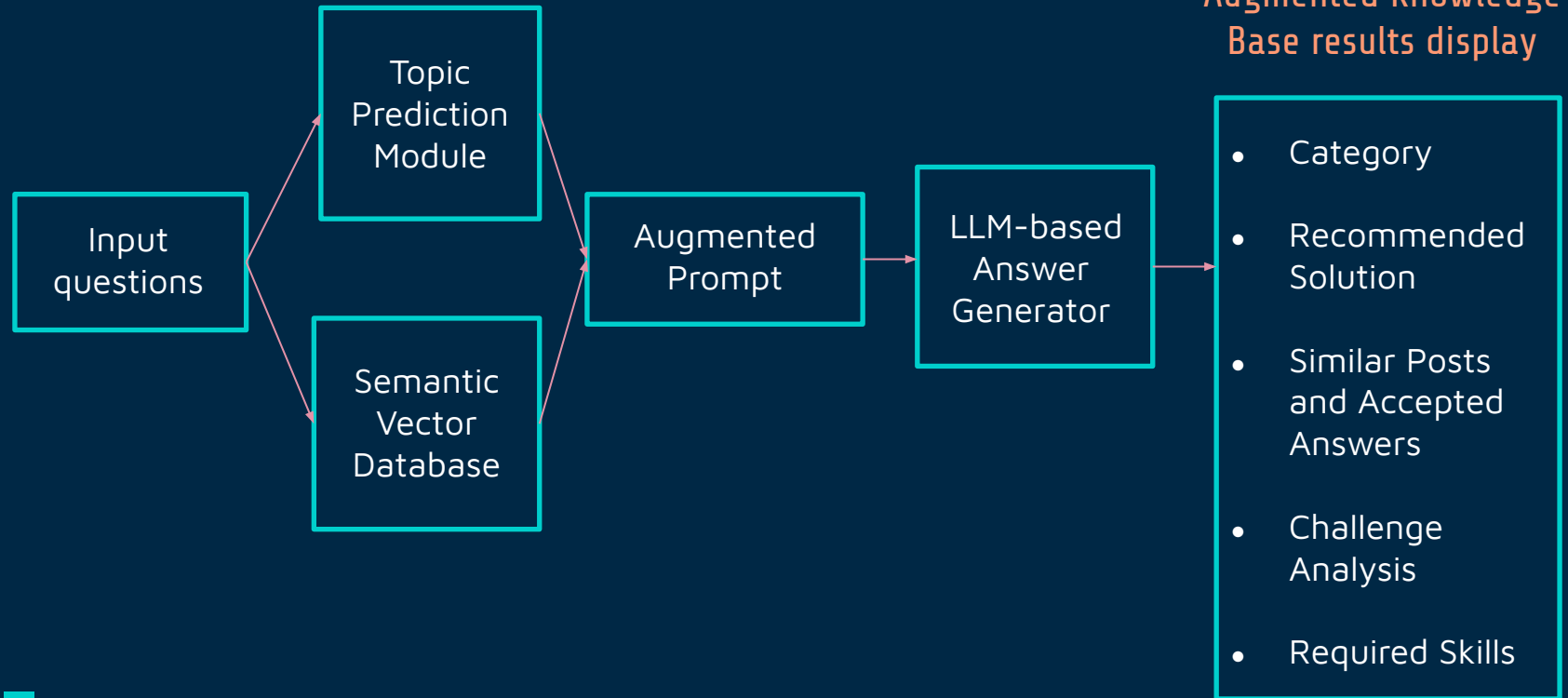
385 questions and accepted answers from stack overflow.

Supporting more updated data in the future.

1 Introduction



1 Introduction



Background

02

Economic Losses

Apache Struts Vulnerability
\$700 Million Losses

Equifax

2017

2019

Capital One

AWS Misconfiguration
\$80 Million Losses

Unencrypted Cloud Storage
\$80 Million Losses

Apple

2022

2023

Cloudflare

API key exposure
\$80 Million Losses

Challenges Faced by Developers



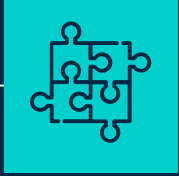
Too Many Programming Languages

Too Much Security Knowledge



Challenges Faced by Developers

Traditional theoretical learning



01

Difficult to
Determine which
Content is Actually
Being Discussed



02

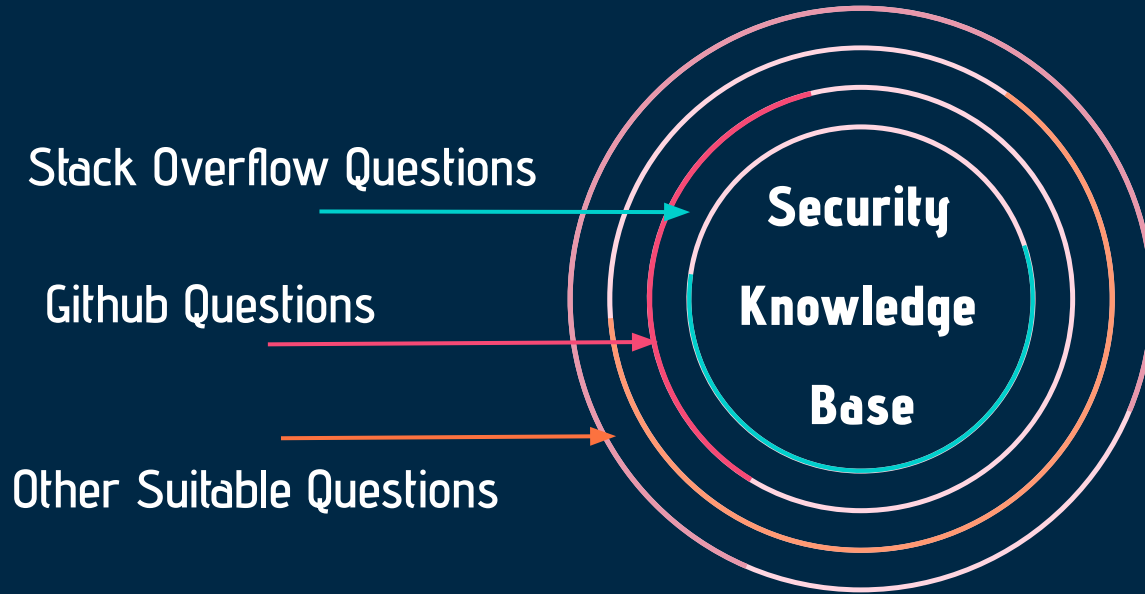
Lacks Best Practices

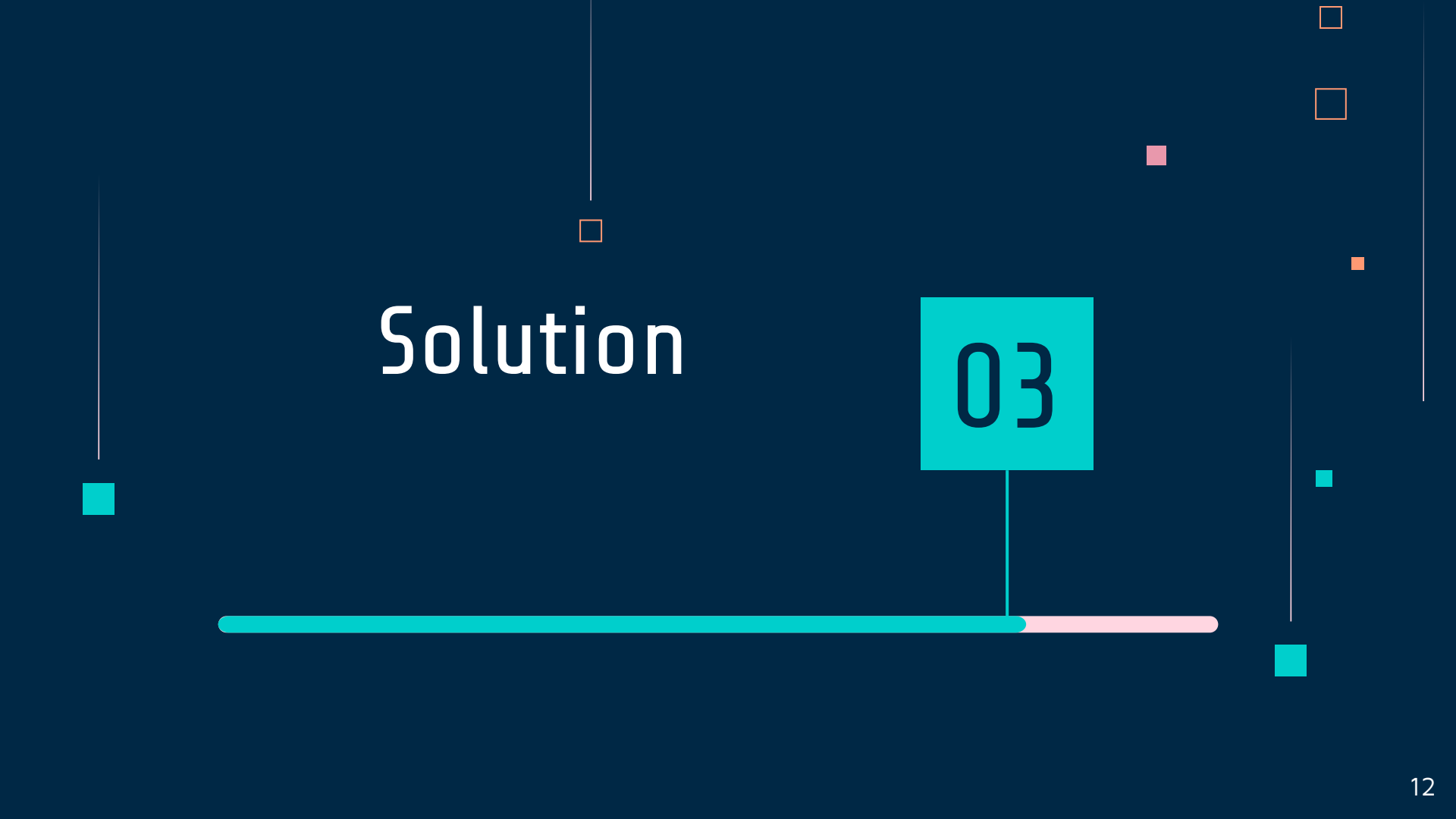


03

Lacks Cross-Domain
Connections

Bridging Learning Gap

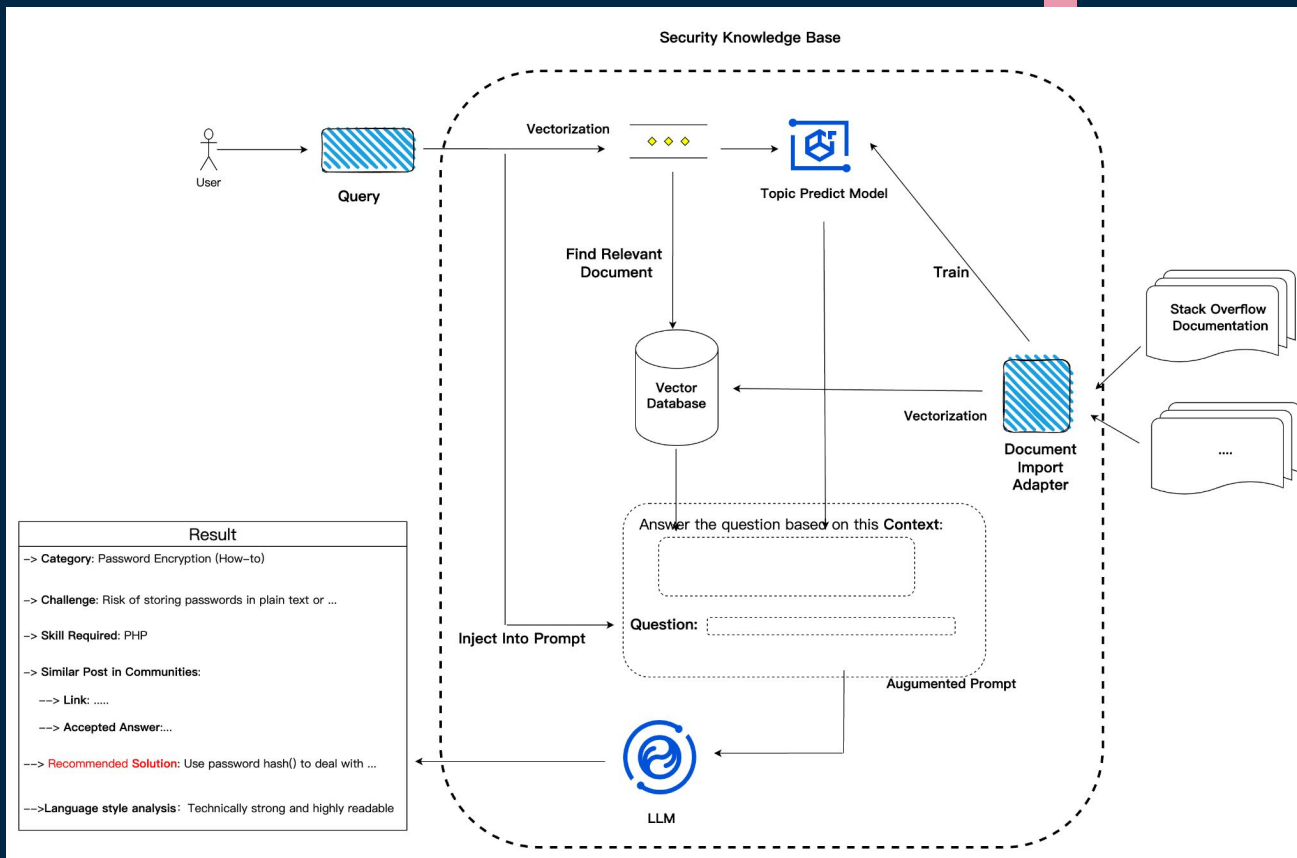




Solution

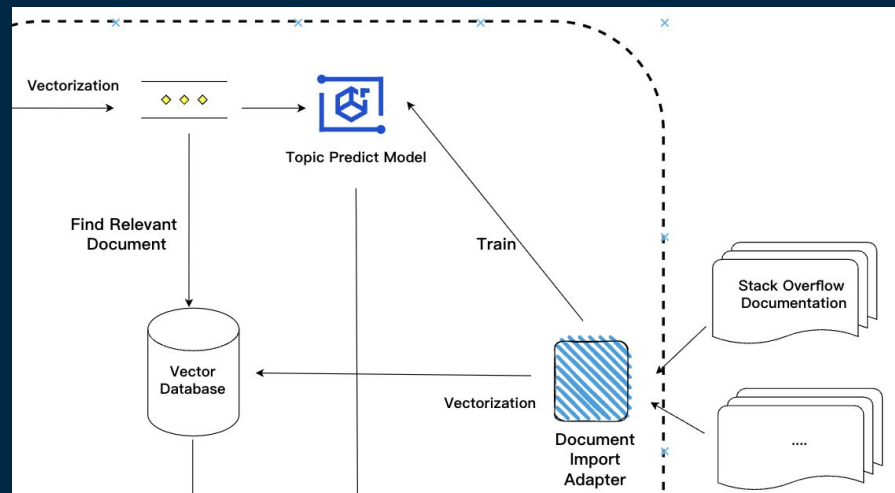
03

Knowledge Base System Design Overview



What we have done

- Vectorize documentations to Vector Database
- Train Topic Predict Models



What we have done

Technic we currently use to train the topic discovery model

BERTopic:



LDA:

Topic ID	Top Keywords (Partial)	Topic Name (Manual Annotation)
0	user, password, code, secur, work	User Authentication and Password Security
1	firebase, mail, keycloak, protocol	Email and Identity Verification Service Integration
2	throw, perform, sandbox, nest	Sandbox Mechanisms and Execution Anomalies
3	salt, rainbow, programm, pseudorandom	Encryption Algorithms and Randomness
4	sql, query, extern, inject, argument	SQL Injection and Web Parameter Processing
5	html, cookie, letter, direct, recommend	Form Handling and User Input
6	python, script, prompt, .net	Python Scripts and Command Execution
7	login, password, user, log, form	Login Systems and User Interface Interaction
8	array, map, assign, result	Data Structures and Array Processing
9	ssl, certif, client, http, connect	SSL Certificates and Client-side Connections
10	keystore, edit, key, modifi, detect	Key Storage and Edit Operations
11	encrypt, decrypt, java, php	Symmetric/Asymmetric Encryption Logic
12	soap, wsdl, servic, goe, why	SOAP Interfaces and Service Communication
13	error, insert, guid, debug	Console Debugging and Form Validation
14	stackoverflow, ssh, putti, bounty	Command Line Tools and Stack Overflow Citation

What we found:

- Relationship between different security knowledge words

(Example : **Spring Framework-based Java** Application have more issues related to **user authentication**)

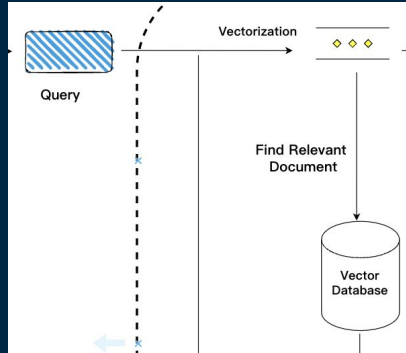
- Some new topics we ignored

What we do next:

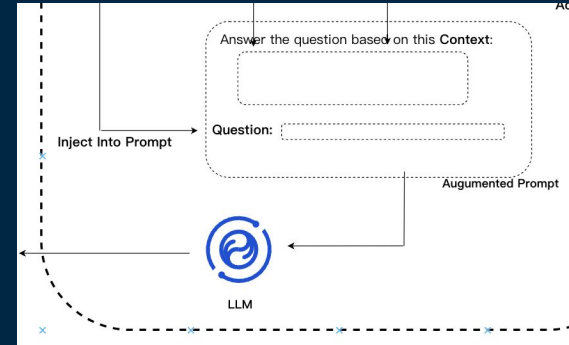
- Comparing these two models
- Test and decide which model to use

Next Step

1. Find relevant Results - Using cosine similarity



2. LLM - local deploy



3. Test:

- Category: Supervised learning and unsupervised learning
Compare different category prediction model(LDA / BERTopic)
- Compare different vectorize methods
- Compare LLM generated results
- Record and analyse the test results

Key Benefits of the Knowledge Base

04

Key Benefits of the Knowledge Base

Capable of importing multiple types of security documents

01

Example

- Capable of importing other security documents source, such as github or <https://www.cve.org/> and etc.

Capable of updating external security documents to keep freshness.

02

Example

- GPT-4o's database update ends in June 2024
- Our knowledge base capable of importing security knowledge update to 2025

Next Plans & Conclusion

05

Next Plans

ROADMAPS			
Preprocessing on Stack Overflow data	Week 1	Record and analyse the results	Week 7
Extract challenges and required skills	Week 2	Milestone 1 - Product Demo Delivery	Week 8
Supervised learning(SVM) & Unsupervised Learning	Week 3	Testing Plan and prepare final testing	Week 9
Pitch Presentation	Week 4	Debugging & Bug fix	Week 10
Build up the pipeline	Week 5	Product Testing	Week 11
Import documentation and run the pipeline	Week 6	Final Product Delivery & Report	Week 12

Conclusion

- Security breaches remain a costly and pressing issue.
- Developers need faster, smarter access to practical security knowledge.
- Our AI-driven system provides structured, real-world solutions from real-world data.
- Our MVP will be completed by the end of this semester and will guide developers to better security practices, faster.

The background is a dark blue gradient. It is decorated with various geometric elements: thin white vertical lines of varying lengths, small squares in teal, orange, and pink, and larger squares in teal and orange. The text 'Any Questions?' is centered on the slide. 'Any' is in white and 'Questions?' is in orange.

Any
Questions?